

Practical Real-time CV-QKD with a Chip-scale Receiver

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Abstract Scalable hardware and real-time operation are key requirements for field-deployable CV-QKD. We demonstrate an automated CV-QKD system featuring a miniaturized receiver, built on chip-scale telecom laser and integrated coherent receiver. This system enables real-time key generation over 26 km and 52 km links. ©2026 The Author(s)

Introduction

The development of quantum computers represents a growing threat to classical cryptography, as algorithms like Shor's [1] could break widely adopted asymmetric cryptography schemes. This endangers present and future communications, since all stored encrypted data could be decrypted in the future. Quantum Key Distribution (QKD) changes the paradigm, deriving its security from quantum properties of light instead of mathematical algorithms. Among QKD protocols, Continuous Variable QKD (CV-QKD) [2] is a promising approach, since it encodes information on the quadratures of light, making it inherently compatible with classical coherent communications. This allows leveraging established telecom technology and simplifies integration into existing networks. Similar to coherent communications, CV-QKD systems can leverage photonic integration to miniaturize components, reduce cost and complexity, and enable mass manufacturing. This represents a great advantage for a wide deployment of QKD. Several studies have demonstrated the use of chip-scale components [3–14], showing the potential of integrated photonics for CV-QKD scalability. Focusing on receiver miniaturization, previous studies showed CV-QKD with chip-scale receivers [3, 4, 6–9, 11–13]. However, these implementations are confined to laboratory settings. These receivers still rely on discrete, expensive, or impractical components, like bulky ultra-low-linewidth local oscillator (LO) lasers, high-bandwidth oscilloscopes, or manual polarization controllers, preventing the realization of complete, deployable chip-scale modules. Moreover, these studies relied on offline post-processing, which made the systems unable to operate in real applications, where key updates for real-time encryption should be automated and continuous. In this work, we overcome these limitations by demonstrating a complete chip-scale receiver module built using chip-scale telecom modules, enabled by a micro-integrated external cavity (EC) laser and a dual-polarization integrated coherent receiver (ICR). The miniaturized receiver is

integrated into an automated CV-QKD architecture, allowing for real-time key generation.

System

The system builds on the commercial CV-QKD system Andariel, developed by KEEQuant GmbH, with the receiver's optical components upgraded to chip-scale counterparts. A schematic of this setup, including a close-up of the chip-scale modules, is shown in Fig. 1. The system is composed of two separate subsystems, the transmitter (TX) and the receiver (RX), which are connected by a single optical fiber (i.e. the quantum channel). The TX consists of discrete optical components, including a low-linewidth tunable laser (<10 kHz), an IQ modulator, a VOA, and a photodiode. Custom-made electronics allow for the control of all components and the generation of the payload through a digital-to-analog converter, driving the modulator at a symbol rate (f_{sym}) of 156.25 MBaud. The miniaturized receiver is composed of a chip-scale laser, serving as the local LO, and an integrated coherent receiver (ICR), alongside two optical switches used for the calibration of the shot noise unit. The laser is a micro-integrated external cavity laser consisting of an InP gain chip coupled to a micro-optical external cavity, with linewidth below 30 kHz. The ICR is based on a monolithic SiPh integrated circuit, composed by a dual-polarization optical hybrid, balanced photodiodes, and a co-integrated transimpedance amplifier (TIA). Both modules are industrial-grade telecom components leveraged for CV-QKD applications. Custom-designed electronics allow for the control of the various RX components and the digitization of the received signal through an analog-to-digital converter. The TX and RX CPUs implement the automated protocol pipeline, including post-processing. Compared to the discrete optics implementation of the receiver, which includes an optical hybrid, a discrete low-linewidth tunable laser, and a balanced detector with a low-noise TIA, we are able to reduce the size of the optics to 6% of the original footprint and 2% of the original volume. The system implements the

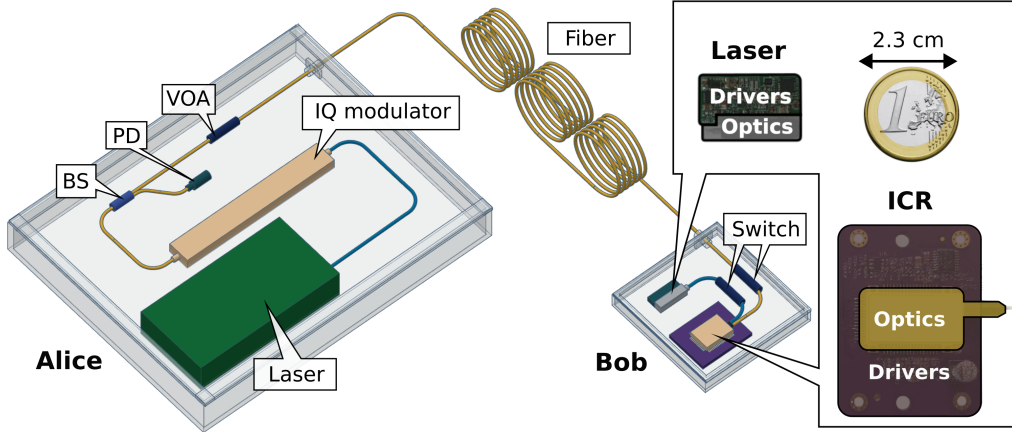


Fig. 1: Experimental setup of the CV-QKD system, with the discrete transmitter on the left and the chip-scale receiver on the right, with pictures of the presented modules (partially obfuscated), to scale.

Gaussian-modulated (GM), prepare-and-measure, no-switching protocol [15]. The security proof assumes the asymptotic regime, individual (or collective) attacks, and trusted detector [16, 17]. The secret key fraction (SKF) is then given by

$$\text{SKF} = (1 - \nu)(1 - \text{FER}) (\beta I_{AB} - I_{BE}), \quad (1)$$

where I_{AB} denotes the mutual information, and I_{BE} represents Eve's Shannon information under reverse reconciliation. For collective attacks, I_{BE} is replaced by the Holevo bound χ_{BE} . Moreover, β is the reconciliation efficiency, FER is the frame error rate, and ν is the fraction of disclosed symbols. The protocol is fully automated and executed a series of shots (i.e. protocol instances) that include system tuning, shot noise unit (SNU) calibration, quantum states exchange, parameter estimation, post-processing, key confirmation, and privacy amplification. The digital signal processing (DSP) chain is separated between TX and RX and executed by the CPUs. On the TX side, it includes a pre-emphasis filter, pulse shaping, up-sampling, upmixing, and multiplexing with a pilot tone. On the RX side, it includes frequency offset compensation, channel equalization, polarization and phase recovery via pilot tone, downmixing, and downsampling. Clock and LO frequency synchronization is performed over the quantum channel using optically modulated tones.

Results

The system featuring the chip-scale receiver was evaluated over optical fiber spools of 26 km and 52 km during 24-hour measurements. Operating autonomously, it generated keys in real time, which were stored in a key buffer and made available for encryption. The real-time operation of the system is shown in Fig. 2. Fig. 3 shows the experimental and estimated SKF for both individual and collective attacks, with individual attacks assumed during post-processing. The overall CV-QKD parameters are reported in Table 2. The system performed

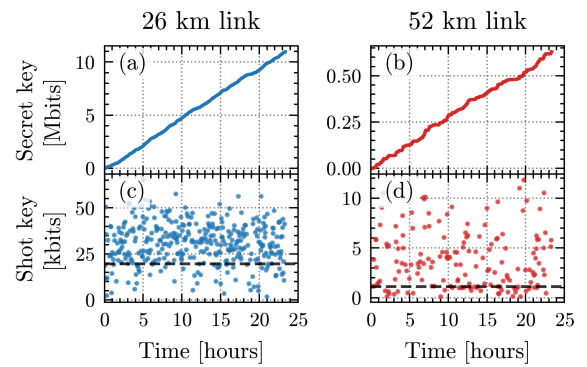


Fig. 2: Real-time key generation over fiber links. (a),(b) report the key length over time, while (c),(d) report the key length per successful shot. The dashed black lines represent the average key length per shot, also accounting for unsuccessful shots.

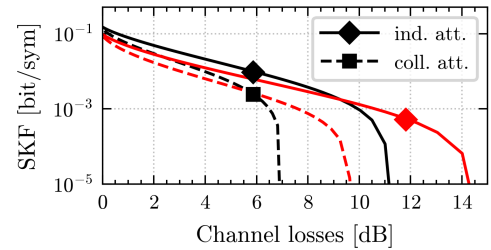


Fig. 3: SKF vs. channel losses for 26 km (black) and 52 km (red) fibers under individual and collective attacks. Markers denote measured SKFs; lines represent estimates, assuming constant excess noise at the channel output.

558 and 572 shots during the evaluations, generating 10.9 and 0.6 Mbits of secret key, respectively. Key updates were frequent, guaranteeing reliable operation for real-world applications. The success rates of the protocol were 64% and 29%. During operation, the resulting real-time key rates (SKR_{rt}) were 133 and 8 bit/s. In terms of quantum-states exchange key rate ($\text{SKR}_{\text{qse}} = \text{SKF} \cdot f_{\text{sym}}$), typical metric of offline demonstrations and defined according to [14], we obtain 1.46 Mbit/s and 82 kbit/s.

Tab. 1: CV-QKD receiver parameters, including efficiency, noise and QCNR. The QCNR considers LO power of 20 mW.

η_{rec}	ξ_{rec} [SNU]	QCNR [dB]
0.30	0.2	10.0

Tab. 2: Experimental CV-QKD parameters over 26 and 52 km optical fibers, including: channel length, real-time and quantum states exchange key rates, channel excess noise, channel transmittance, modulation variance and shot success rate.

L_{ch} [km]	SKR_{rt} [bit/s]	SKR_{qse} [Mbit/s]	ξ_{ch} [mSNU]	T_{ch} [dB]	V_{mod} [SNU]	P_{succ}
26	133	1.46	19.7	-5.87	3.6	63.8%
52	8	0.082	8.8	-11.81	6.0	28.7%

Tab. 3: Comparison of the CV-QKD experimental demonstrations with chip-scale receivers over optical fibers. The reported data include the symbol rate, channel length, real-time and quantum states exchange key rates, single (SP) or dual (DP) polarization receiver, type of LO laser, discrete (DM) or gaussian (GM) modulation, security proof (I.A./C.A.: individual/collective attacks).

Ref.	f_{sym} [Mbaud]	L_{ch} [km]	SKR_{rt} [bit/s]	SKR_{qse} [Mbit/s]	Pol.	Laser	Mod.	Proof
[3]	0.8	0.002	-	0.25	SP	Bulk*	GM	I.A. [†]
[4]	250	50 / 100	-	0.75 / 0.0007	SP	Chip	GM	C.A. [†]
[6]	1000	28.6	-	0.24	SP	Bulk	GM	C.A. [‡]
[7]	500	20 to 100	-	10.37 to 0.06	SP	Bulk	GM	C.A. [†]
[8]	10000	5 / 10	-	0.737 / 351	SP	Bulk	DM	C.A. [‡]
[9]	250	25.2	-	1.9	DP	Bulk	DM	C.A. [‡]
[11]	16000	20	-	246	SP	Bulk	DM	C.A. [‡]
[12]	40000	10	-	1213	DP	Bulk	DM	C.A. ^{‡,§}
[13]	1562.5	25.8 / 50.4	-	31.05 / 5.05	SP	Bulk	GM	C.A. [†]
Ours	156.25	26 / 52	133 / 8	1.46 / 0.08	DP	Chip	GM	I.A. [†]

* Transmitted LO, [†] Asymptotic regime, [‡] Finite-size regime, [§] Composable security

The significant difference with the SKR_{rt} arises from the computational load of the low-density parity-check (LDPC) codes, which currently represent the system bottleneck. The average excess noise of the system at the channel output, considering the successful shots, was 19.7 and 8.8 mSNU, with channel losses of 5.9 and 11.8 dB. The receiver performance, summarized in Table 1, exhibits an electronic noise (ξ_{rec}) of 0.2 SNU and an efficiency (η_{rec}) of 0.3. Notably, the quantum-to-classical noise ratio (QCNr) reaches 10 dB, given 20 mW of input LO power.

Discussion

This study follows prior demonstrations of CV-QKD with chip-scale receivers, while additionally addressing the commercial feasibility of the architecture. To contextualize our work, Table 3 compares our results with previous experimental demonstrations. Notably, because our system automates the entire protocol, we are able to report the real-time key rate. Unlike prior studies that rely on offline post-processing and therefore omit this parameter, providing a real-time key rate represents an important step toward the practical usability of CV-QKD in real deployment scenarios. In terms of hardware, our approach overcomes key scalability limitations of previous designs. While most prior studies required discrete ultra-low phase noise LO lasers, our system avoids this constraint by utilizing a chip-scale alternative. Furthermore, we eliminate the need for single-polarization receivers with bulky, manually adjusted polarization controllers, thereby improving system deployability and resilience to polarization drift. Although our symbol rate is lower than in several existing systems, this is a deliberate choice. Because the

current bottleneck is the post-processing rate, increasing the symbol rate would unnecessarily inflate hardware costs without yielding meaningful performance improvements. In this work, we assume individual rather than collective attacks as a practical trade-off for industrial deployments. This maximizes the key rate while maintaining practical security, since the technology required for collective measurements remains far beyond current capabilities. The system can nonetheless operate assuming collective attacks (Fig. 3), with reduced performance. Finite-size effects and composable security will be addressed in future work. By utilizing mature, miniaturized commercial telecom components co-integrated with electronics, we significantly reduce the development overhead, accelerating the path toward scalable, industrial-grade QKD. Ultimately, by demonstrating secret key generation over 26 km and 52 km channels, we prove that this solution is immediately viable for coverage of metropolitan-scale links, with the potential for further expansion using trusted nodes.

Conclusions

This work demonstrates a practical CV-QKD system featuring a chip-scale receiver operating in real time. Advancing beyond typical laboratory demonstrations, the system automates the entire protocol pipeline (including online post-processing) and utilizes miniaturized, deployable hardware based on mature commercial telecom technology. These characteristics enable a scalable solution for practical CV-QKD networks. Future research will focus on integrating both transmitter and receiver to achieve a fully scalable hardware architecture, as well as optimizing the post-processing pipeline to further increase real-time key rates.

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